Tertiary Infotech Academy Pte Ltd

Lab 11 — Run a Finance Data-Quality Audit

Python Programming for Finance · TGS-2026065049 · LO5

Scenario

A credit portfolio extract must pass completeness, uniqueness, range and reconciliation controls before modelling.

Goal

A reusable audit function and exception report for loan records.

Tools

Pandas, CSV, JSON

Step-by-step

1. Load loans_dirty.csv and define the audit controls.

2. Check required columns and row uniqueness.

3. Measure missingness by column.

4. Test ranges for income, debt-to-income and credit score.

```
python starter.py
```

5. Write exceptions.csv and audit_summary.json.

```
python solution.py
```

6. Confirm exception counts reconcile to the summary.

```
python -m unittest test_solution.py
```

Acceptance criteria

The audit identifies every planted defect, produces machine-readable counts and leaves the raw file unchanged.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.